NEEDLE: A Live Benchmark of Web Search APIs for AI Agents

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Abstract

NEEDLE is a live benchmark of web search APIs for AI agents. Agent benchmarks rarely test the search API alone; NEEDLE scores what the API returns to the agent: titles, URLs and snippets. Hourly and daily runs draw new queries from public sources for news, finance, papers and law, plus a fixed stream of rare-word queries, and compare each API with the best results any tested API found. Over eight weeks we test 26 API settings (engines) from 20 vendors. On 12.4k shared query-runs, paired tests order 178 of 210 engine pairs. Title lookups hide large gaps: 23 engines find a paper from its title at recall@10 .71–.98, but from body keywords at only .08–.72. These results provide evidence that API comparisons should also test queries built from a paper's body.

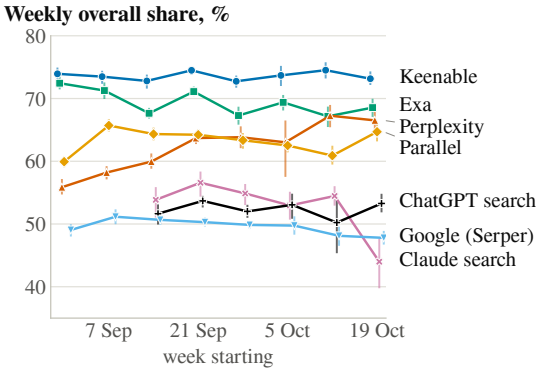


Figure 1: The top rank holds all eight weeks; lower ranks change. Overall share by ISO week, 31 August–25 October 2026. ChatGPT and Claude search start on 14 September; the pool grows from 18 to 25 engines. Points are query-weighted means; bars are 95% run-level bootstrap intervals. Scores follow the dashboard rules.

1 Introduction

An AI agent may need the paper behind a detail it read elsewhere, such as the query "gap rent regularity" (Table 2), whose words are absent from the paper's title and abstract. A web search API decides which sources the agent can inspect, and shows only a title, URL and snippet before the agent opens a page.

Testing the API alone is hard. The live web has no fixed corpus or complete relevance labels; fixed test sets go stale and can leak into training data (Sainz et al., 2023; Balloccu et al., 2024); and a score must use what the API returns, not the page behind it. End-to-end agent benchmarks do not isolate the API (Section 2).

NEEDLE tests this step directly: one query in, up to k results out, scored on the evidence returned to the agent, for five tasks with new queries each run. Scorers compare each engine with a pooled ceiling: the best results any tested engine found. Registry facts and rare queries can repeat, and all judge checks use LLMs, not humans. Our claims:

- Paired tests order 178 of 210 pairs of the 21 main engines on 12.4k shared query-runs; 32 pairs stay unresolved (Section 4, Table 3).
- Title lookups hide large gaps: 23 of 26 engines reach scholar recall@10 .71–.98 from titles but .08–.72 from body keywords (Section 4, Figure 2).
- Weekly ranks are stable but not fixed: adjacent weekly rankings agree at Kendall's τ_b .81–.94 (Section 4, Figure 1).

2 Related Work

Live and API benchmarks. LiveBench and LiveSearchBench draw new items from dated sources but score model answers (White et al., 2025; Zhou et al., 2026a); NEEDLE scores the search results behind such answers. BrowseComp scores agents end to end (Wei et al., 2025); FreshStack and BrowseComp-Plus fix the corpus to isolate retrieval (Thakur et al., 2025; Chen et al., 2026), while NEEDLE keeps the live web.

API comparisons use static questions under one agent (Selvam and Ghosh, 2026; Artificial Analysis, 2026), and web engine audits use fixed query sets (Lewandowski, 2015; Kirsten et al., 2026); NEEDLE draws new queries each run and scores the API without an agent.

Scorers and tasks. LLM assessors and simulated known-item queries can reproduce system rankings (Thomas et al., 2024; Upadhyay et al., 2025; Azzopardi et al., 2007; He et al., 2025); NEEDLE uses both and checks its judge against raters from other vendors. Domain benchmarks cover literature, caselaw and finance search on fixed sets (Ajith et al., 2024; Hou et al., 2025; Hu et al., 2026); NEEDLE runs these tasks on new queries each day. Full text helps retrieval beyond abstracts (Lin, 2009); our body queries test that gap on live APIs.

3 The Benchmark

An *engine* is an API at one setting. We test 26 engines from 20 vendors, including five mode variants. A *query-run* is one query issued in one run. Appendix A gives the API settings and data sources.

3.1 Queries

Table 1 summarizes the run schedule and scoring metrics.

News. Each hour an LLM writes 20 event queries from news feeds and Google Trends, rejecting evergreen items.

Finance. Queries ask for company facts from Wikidata (Vrandečić and Krötzsch, 2014) and GLEIF, quarterly 10-Q figures from SEC XBRL, or filings by description.

Scholar. Each recent arXiv or Europe PMC (Europe PMC Consortium, 2015) paper yields four paired known-item queries (Azzopardi et al., 2007): its normalized *title*; *body keywords* for a detail absent from title and abstract; a *body question* about that detail; and a *tip-of-tongue* description (Arguello et al., 2021) without names or exact values. All four must pass the filters.

Legal. CourtListener opinions yield caption queries; eCFR sections yield queries based on verbatim spans. Filters reject citation leaks.

Rare. We sample a fixed public stream of English agent queries containing rare words (Ning et al., 2026; Li et al., 2026; Zhou et al., 2026b). Table 2 gives query examples; Appendix B gives the filters.

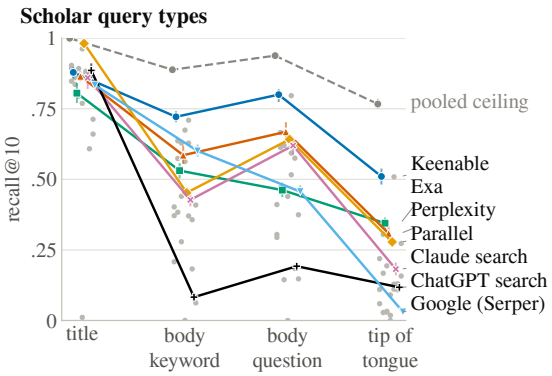


Figure 2: Engines that find papers from titles differ widely on body queries. Scholar recall@10 by query type: 31 runs, 29 September–28 October 2026; 2,600–3,005 query-runs per line and type. Lines show seven main engines, grey dots the other 19 engines, and dashes the pooled ceiling. Points are query-weighted means; bars are 95% run-level bootstrap intervals.

At least half of legal queries and finance money and filing queries carry quotes, `site:`, `after:` or `before:`. The client maps operators to native filters where available.

Task	Cadence	Queries/run	k	Metric
news	hourly	20	5	nDCG@5
finance	daily	120	5	recall@5
scholar	daily	up to 420	10	recall@10
legal	daily	40 + 0–20	5	recall@5
rare	daily	100	5	nDCG@5

Table 1: Benchmark tasks, run frequency and primary metrics. k is the number of search results requested per query. Query counts can fall below the targets after source and query filtering. Legal combines 40 CFR queries with caption queries for recent court opinions.

3.2 Scores

News and rare. GPT-6 rates each distinct result from 0 to 4 using a shared query profile, with checks for relevance, coverage, freshness and literal operator compliance. The scale adapts the Needs Met levels of Google (2025). Ratings 0–4 map to gains $g = \{0, 0, .117, .667, 1\}$, putting high weight on strong results (Kanoulas and Aslam, 2009). Snippets have a 2,000-character cap.

For each query, the *ceiling* keeps the best rating per normalized URL across all engines, variants included, and sorts these ratings (Spärck Jones and van Rijsbergen, 1975; Zobel, 1998). With r_i the rating at rank i, we compute nDCG@5 (Järvelin

Task	Query	Reference answer or profile
news	Paramount Warner Bros takeover closes 2026	closing date; ephemeral event
finance, fact	nvidia number of employees	42,000, ±15%
finance, money	"servicenow" q3 fiscal 2024 operating income	$418M, ±0.1%
scholar, title	optimally selling heterogeneous objects by quantity	arXiv 2610.36244
scholar, body	gap rent regularity	same paper
legal, CFR	"modify or waive any" county committees authority	7 CFR 1417.2
rare	"Krasnogvardeysky District" Russia	judged; rare word has 7 pieces

Table 2: Sample generated queries. News and rare have no reference answer; the judge profile specifies what a result must contain.

and Kekäläinen, 2002) as

$$\mathrm{DCG}_e(q) = \sum_{i=1}^{5} \frac{g(r_i)}{\log_2(i+1)},$$
$$\mathrm{nDCG}_e(q) = \frac{\mathrm{DCG}_e(q)}{\mathrm{IDCG}(q)}.$$

IDCG is the DCG of the top five pooled ratings. Missing ranks contribute zero. Repeated normalized URLs receive zero gain except on `site:` queries.[1] We drop a query-run from all engines if any judgment fails or no result reaches rating 2. The task score is the mean over retained query-runs; the judged ceiling has mean 1.

For example, A returns ratings 4 and 2; B returns 3 and 4, with the 4-rated URL shared. The pool contains 4, 3, 2, so $\mathrm{IDCG} = 1 + .667/\log_2 3 + .117/2 = 1.479$. A scores .73 and B .88: the denominator uses the shared pool, not each engine's own best ordering.

Scholar, legal and finance. Recall@k counts queries with a matching result in the top k. Scholar matches arXiv ids, DOIs and PMIDs; legal matches citations, dockets and identifiers in URLs and snippets. Caselaw recall measures case identity, not necessarily the opinion page. Finance checks whether the snippets contain the answer, with a 0.1% money band and a 15% headcount band. The matcher reads table figures and parenthesized losses. An LLM may upgrade a non-money miss, but cannot revoke a hit; a failed backstop judgment excludes an engine's query only if no result matches. The ceiling is a hit by any engine. Appendix F checks matcher error.

[1] The pool still keeps each URL once, so this exception can yield nDCG@5 above 1. The exception affects 3 of 1,748 rare `site:` engine–query pairs in the main window.

3.3 Overall share and failures

For task metric M_t, an engine's overall share is

$$S(e) = \frac{1}{5} \sum_{t=1}^{5} \frac{M_t(e)}{M_t(\text{ceiling})}.$$

The tasks have equal weight. The ceiling is the best the tested engines found, not the best the web contains.

Table 3 uses dashboard scores from the 276 runs in which all 21 main engines took part. On finance, scholar and legal, we exclude account-side failures (our quota, rate limit, login or credit); other failures score zero. On news and rare, every failure in this window scores zero. These exclusions remove up to 28% of an engine's query-runs.

Comparing two engines. Query difficulty varies widely and affects all engines alike, so individual intervals can overlap even when one engine beats the other on the same queries. We therefore estimate the difference $S(a) - S(b)$ on shared runs. Within each task, we sample whole runs with replacement 2,000 times, drawing the same runs for every engine (Efron and Tibshirani, 1993). Each sample preserves within-run dependence and each engine's exclusions; we recompute query-weighted task and ceiling means, then overall shares. We use the bootstrap standard error to form a normal-approximation interval for each difference. Bonferroni correction accounts for testing all $\binom{21}{2} = 210$ pairs at once, at a 5% familywise level (Carterette, 2012). An interval that excludes zero supports ordering that pair under these scoring rules. The intervals measure sampling uncertainty, not scorer error. Appendix C gives the interval formula and checks other sampling units and failure policies.

3.4 Judge validation

We sample 100 news and 100 rare queries, with six distinct top-five URLs per query (1,200 results).

Engine	Overall share, %		Scholar	News	Finance	Rare	Legal	Failed, %	Latency
	mean	95% CI	R@10	nDCG@5	R@5	nDCG@5	R@5	requests	p_{50}, ms
ceiling (pooled)	*100*		*.901*	*1*	*.993*	*1*	*.971*		
Keenable	73.2	72.3–74.1	**.731**	.609	.903	**.508**	**.797**	0.0	135
Exa	68.5	67.4–69.4	.563	.599	**.922**	.476	.771	0.3	2,132
Perplexity	66.9	65.3–68.5	.578	.640	.895	.468	.674	0.0	433
Parallel	64.9	63.6–66.1	.587	.535	.907	.387	.734	0.8	2,660
Brave (LLM ctx)	64.6	63.4–65.6	.533	**.642**	.858	.454	.658	0.0	1,112
Linkup	57.6	56.1–58.9	.470	.567	.797	.400	.570	0.0	1,212
Brave	57.3	56.2–58.4	.488	.557	.769	.354	.621	0.0	487
Context	57.1	55.4–58.6	.598	.469	.782	.339	.576	0.0	866
DeepSeek	56.9	55.8–57.9	.271	.635	.818	.387	.676	0.0	551
You.com	55.8	54.7–56.9	.471	.553	.747	.343	.602	0.0	486
ChatGPT search†§	51.5	49.4–53.1	.300	.340	.809	.366	.697	11.1	43,831
Kagi§	51.3	49.9–52.6	.560	.301	.728	.275	.617	7.6	1,275
Firecrawl	50.7	49.2–52.0	.488	.286	.820	.265	.598	0.0	688
Octen	50.2	48.8–51.7	.307	.511	.730	.322	.586	0.0	197
Google (Serper)	47.7	46.8–48.6	.475	.262	.738	.255	.582	0.0	1,785
Claude search†§	45.1	42.7–47.6	.498	.276	.685	.195	.525	35.9	33,906
TinyFish§	44.5	43.0–45.9	.429	.256	.730	.220	.521	29.0	1,904
Bing (SearchAPI)	44.1	41.1–46.8	.416	.288	.675	.173	.586	0.4	2,008
Jina	43.8	42.4–45.1	.492	.210	.726	.163	.523	4.9	17,272
Tavily	42.6	41.5–43.7	.303	.303	.828	.284	.361	0.1	1,999
Ceramic	14.7	13.9–15.7	.001	.040	.441	.109	.137	0.1	99

Table 3: Results on 276 shared runs, 17–28 October 2026 (12.4k query-runs, 21 engines); the ceiling also includes variants. Task cells are absolute metrics, best in bold; italics mark the ceiling. Failed counts all request failures. Latency is the per-run median on news. †Server-side search tool; latency includes answer generation. §Account-side exclusions leave rank uncertain (Appendix C).

Two Claude Opus 5.5 raters, Gemini 4 Argon and Grok 4.7 grade these with engine names hidden and a common written guideline. Claude raters also see the first 3,000 page characters. The Claude raters agree at quadratic-weighted $\kappa = .94$ (Cohen, 1968); Gemini and Grok agree with their consensus at .92 and .90. Agreement with the production judge is .77–.79 (Table 4). A separate snippet-only pass on 400 items finds that adding page text changes 25% of grades.

To test rankings on the same evidence as production, Gemini and Grok also grade 100 complete query pools (50 per task): 5,698 unique results from all 21 engines, using snippets only. Each gives engine-rank Spearman .94 against the production judge (95% intervals .89–.97 and .87–.96). All three retain the same top five in different orders; none separates Keenable from Exa on this sample. These checks measure model agreement, not human validation. Appendix E gives the audit protocol and checks sensitivity to evidence length and literal quotes.

Quadratic-weighted κ	News	Rare	All
Claude A vs. B	.94	.94	.94
Judge vs. Claude consensus	.72	.80	.77
Gemini vs. consensus	.91	.92	.92
Grok vs. consensus	.90	.88	.90
Gemini vs. judge	.74	.81	.78
Grok vs. judge	.75	.81	.79

Table 4: Agreement on 1,200 blinded results, 600 per task. Consensus is the rounded mean of the two Claude page grades. The judge–consensus 95% query-bootstrap interval is .74–.81.

4 Results

The data support most pairwise rankings, not every rank. The adjusted score-difference intervals exclude zero for 178 of 210 engine pairs. We cannot reliably order the remaining 32 pairs from this sample; an unresolved pair does not mean equal scores. The overall lead also depends on the task mix: scholar contributes 3.7 points of Keenable's 4.7-point lead over Exa. Without scholar, the lead is 1.2 points (Appendix C).

Body queries expose gaps that title queries hide. Over 31 scholar runs, 23 of 26 engines reach recall@10 .71–.98 from titles. The same 23 span .08–.72 on body keywords (Figure 2; Table 7). Keenable leads at .72, followed by its realtime mode (.71) and Kagi (.67). Appendix D gives paired intervals on common papers.

Weekly ranks are stable, not fixed. Adjacent weekly rankings agree at Kendall's τ_b (Kendall, 1945) .81–.94. Keenable leads all eight weeks, but lower ranks change (Figure 1). Appendix C checks changes in the engine pool, account failures and dispatch times.

Limitations

Conflict of interest. The authors' organization builds an evaluated engine. The code, prompts, clients and per-query outputs are public so readers can audit the claims. All engines receive the same queries, snippet cap, request pacing and judge; each keeps one setting throughout the main window. We cannot tell whether a vendor changed its index for the benchmark or for other reasons.

Evidence and judge. Scores measure the API response, not the fetched page. Short snippets can hide relevant content (Turpin et al., 2009; Kazai et al., 2022), and identifier matching can miss a target without its id. On 400 sampled results, page text changes 25% of grades. News and rare depend on one judge; all audit raters are LLMs, and none provides human validation, which LLM labels cannot replace (Faggioli et al., 2023; Clarke and Dietz, 2025). Shared rules can cause shared errors. Query-term stuffing can fool judges (Alaofi et al., 2024); vendor and model biases can affect scores (Balog et al., 2025; Soboroff, 2025). The judge shares a vendor with ChatGPT search, and some audit raters share one with Claude search; LLM evaluators can favor their own outputs (Panickssery et al., 2024).

Matcher error. Finance table figures can match the wrong period when snippets omit column headers. In a fresh LLM audit, 5 of 335 table-reading hits contradict the answer; 95 show the right figure without a period label, which the snippet alone cannot confirm. The pooled audit cannot show each engine's error rate. Caselaw recall credits pages about the case, not just the opinion. Appendix F gives the finance checks.

Coverage and weights. Queries are English and US-centric. Equal task weights do not estimate agent traffic, and the lead depends strongly on scholar. Default API tiers need not be each vendor's best. ChatGPT and Claude search return sources through answer-generating tools, so their quality scores are comparable but their latency and cost are not. The ceiling depends on the engine pool (Buckley et al., 2007); compare shares within a common window.

Failures and timing. Excluding account-side failures assumes those requests would score like the rest. Without that assumption, several middle ranks remain uncertain. News and rare also count our credit lapses as zeros in the main window. Engines process queries at different speeds, so slow engines see news later. A five-minute dispatch filter leaves the news order unchanged. Vendor redirect links can break URL matching.

Repetition and reproducibility. Registry facts and the fixed rare stream can repeat. Public sources and queries permit tuning or special handling of benchmark traffic. The studied runs archive queries, reference answers, results and judgments, but lack API publication dates and each run's code revision. Stored ratings reproduce the tables; re-judging cannot exactly reproduce the inputs. Appendix A gives the source terms.

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A API Settings and Data Sources

All engines receive the same query and requested result count. Each engine processes one request at a time; engines run concurrently. Requests have a 30-second timeout and up to three attempts, with exponential back-off from 1 to 30 seconds on HTTP 429, 5xx and timeouts. Short result lists score as returned. Snippets have a 2,000-character cap.

The generators and judge use GPT-6 Luna (`openai/gpt-6-luna`) through OpenRouter, temperature 0 and high reasoning effort. Table 5 lists the search clients and settings. Appendix G gives all prompts. Supporting scripts, samples and labels are under `paper/analysis/`; the manifest identifies each analysis's inputs.

The rare stream draws from DeepResearchGym (Coelho et al., 2026; Ning et al., 2026) (59%, CC BY 4.0), OpenResearcher (Li et al., 2026) (39%, MIT) and LRAT-Train (Zhou et al., 2026b) (1%, no stated license). Source terms govern query texts; our code is MIT and our CC BY 4.0 covers results, judgments and metadata, not source text. LRAT contributes 17 of 1,200 main-window rare queries; removing them changes no mean rare nDCG@5 by more than .003. Queries can contain personal information. Release contacts accept private removal requests; API output records remain subject to vendor terms.

B Query Filters

Scholar samples five domains (computer, physical, life, social and health sciences) and three age bands (7 days, 30 days, 1 year), targeting seven accepted papers per cell. Each paper must pass all four filters:

- **Title:** at least four words, plus a digit, acronym, hyphenated term or five content tokens.
- **Body keywords:** at least three words and two content tokens absent from title and abstract.
- **Body question:** at least eight words and the same novelty test as body keywords.
- **Tip-of-tongue:** at least twelve words; no quotes, decimal, five-digit number or coined title/abstract term; at most two content tokens shared with the title.

An instrumented run accepts 323 of 572 candidate papers (56%). The effect of rejecting other papers on engine order is untested.

News draws from 124 feeds and Google Trends RSS for 52 US geographies; prompts reject evergreen items and request two-to-six-word event queries. Legal captions come from 14 federal courts on CourtListener. CFR queries contain one verbatim two-to-eight-word span but no citation or target section number. Only the quoted variant retains quotation marks. Rare queries must pass fastText English language ID (Joulin et al., 2017, 2016) and contain a word split into at least five BERT WordPieces (Wu et al., 2016; Devlin et al., 2019).

C Statistical Checks and Ranking Sensitivity

Paired intervals. The main window has 12 scholar, 232 news, 10 finance, 12 rare and 10 legal runs. For pair (a, b), let $\widehat{\Delta}_{ab} = S(a) - S(b)$ and let s_{ab} be the standard deviation of its 2,000 bootstrap differences. The adjusted interval is

$$\widehat{\Delta}_{ab} \pm 3.683\, s_{ab}.$$

The interval is a normal approximation, not a percentile bootstrap interval. The implemented critical value is slightly more conservative than 3.675, the two-sided Bonferroni value for 210 comparisons at familywise level .05. Individual engine intervals in Table 3 instead use the 2.5th and 97.5th bootstrap percentiles. Their overlap is not a test of the difference between two engines.

Dependence and missing requests. We repeat the comparisons by resampling days, and by clustering exact query text (target paper for scholar) across runs. The latter checks repeated targets, which day resampling alone does not address. We also restrict the data to query-runs with no account-side failure in any main engine. Of the 178 pairs whose primary adjusted intervals exclude zero, 164 also exclude zero under both day resampling and this matched-support check.

Table 6 separates the failure-policy change from the change in query support. Per-query files cover 274 of the 276 runs; matching retains 4,048 of 12,313 query-runs. With one account-failure policy, Claude search rises from 45.1 to 51.4 overall; matching changes it to 52.6. The matched order has $\tau_b = .89$ against the main table, with Keenable and Exa still first and second. Its difference intervals use 1,000 paired run-bootstrap resamples. If excluded requests instead score zero or at the ceiling, ranks 1–10 stay fixed under both extremes, but Claude search and TinyFish can occupy ranks 11–20. Missing requests therefore limit some rank claims.

Vendor	Endpoint	Variant or non-default parameters
Keenable	api.keenable.ai	default; realtime variant
Exa	api.exa.ai	`type=auto`; instant and lite variants
Google	google.serper.dev (Serper)	default
Bing	searchapi.io (SearchAPI)	engine `bing`
Brave	api.search.brave.com	web search; LLM-context endpoint
Parallel	api.parallel.ai	default processor; turbo variant
Perplexity	api.perplexity.ai	search API; fast variant
Tavily	api.tavily.com	default
Linkup	api.linkup.so	default
TinyFish	api.search.tinyfish.ai	default
Octen	api.octen.ai	default
Ceramic	api.ceramic.ai	default
You.com	ydc-index.io	default
Firecrawl	api.firecrawl.dev	search endpoint, default
Kagi	kagi.com/api/v1	default
Jina	s.jina.ai	default
Context	api.context.dev/v1	default
DeepSeek	api.deepseek.com	`/v1/tools/search`, default
Claude search	api.anthropic.com	`claude-sonnet-5`, `web_search`
ChatGPT search	api.openai.com	`gpt-5.5`, `web_search`, `search_context_size=low`

Table 5: Search clients and settings by vendor. Clients live in `src/needle/shared/search/`; clients also implement the native query filters.

Engine	Historical	Consistent policy	Matched	Δ matched	95% CI for Δ
Keenable	73.2	73.2	74.7	+1.6	+0.4, +2.7
Exa	68.5	68.6	67.5	−1.1	−2.3, +0.2
Perplexity	66.9	66.9	66.2	−0.7	−2.5, +1.1
Parallel	64.9	65.1	63.0	−2.1	−3.5, −0.7
Brave (LLM ctx)	64.6	64.6	67.1	+2.5	+1.1, +3.9
Linkup	57.6	57.6	59.3	+1.7	−0.1, +3.8
Brave	57.3	57.3	59.1	+1.8	+0.2, +3.2
Context	57.1	57.1	56.2	−0.9	−3.0, +1.1
DeepSeek	56.9	56.8	58.0	+1.1	−0.6, +2.9
You.com	55.8	55.8	57.4	+1.6	+0.3, +2.9
ChatGPT search	51.5	52.4	51.9	−0.5	−2.0, +0.8
Kagi	51.3	52.1	52.3	+0.1	−0.8, +1.2
Firecrawl	50.7	50.7	50.1	−0.5	−1.9, +0.9
Octen	50.2	50.2	49.1	−1.1	−2.7, +0.6
Google (Serper)	47.7	47.7	47.4	−0.3	−1.5, +0.9
Claude search	45.1	51.4	52.6	+1.2	−0.4, +2.9
TinyFish	44.5	45.1	44.7	−0.4	−2.0, +1.2
Bing (SearchAPI)	44.1	44.1	48.3	+4.1	+1.5, +7.5
Jina	43.8	43.8	43.2	−0.5	−2.2, +1.2
Tavily	42.6	42.6	42.5	−0.1	−1.2, +1.3
Ceramic	14.7	14.7	14.7	−0.1	−1.1, +1.2

Table 6: Overall shares in percent. Historical repeats Table 3. Consistent policy excludes account-side failures on every task and scores other failures zero, using 12,313 available query-runs. Matched restricts these to 4,048 query-runs with no account failure in any main engine. Δ is Matched minus Consistent policy, in percentage points, with a paired run-bootstrap interval.

Normalization, weights and timing. Removing an engine from the pool changes its overall share by at most 0.8 points. Dropping any one task preserves the leader but changes the Keenable–Exa gap from 1.2 to 6.4 points. The lead's size is sensitive to task choice even when the leader stays the same.

Among the 15 engines present in all eight plotted weeks, adjacent rankings have $\tau_b = .85$–$.94$. Fixing the news pool to those 15 changes no sampled weekly order. Restricting news comparisons to requests sent within five minutes of the first engine retains 96% of scored pairs, leaves the order unchanged and moves no mean nDCG@5 by more than .010. These checks address changing engine participation and dispatch lag.

Engine	Title	Body keyword	Body question	Tip of tongue	Runs	Query-runs
ceiling (pooled)	*1*	*.888*	*.938*	*.767*	*31*	*12,020*
Parallel	**.981**	.453	.642	.278	31	12,020
Kagi	.962	.674	.640	.091	30	11,599
Bing (SearchAPI)	.903	.402	.411	.060	31	12,020
Parallel Turbo	.894	.208	.294	.110	31	12,020
Brave (LLM ctx)	.893	.440	.620	.194	31	12,020
ChatGPT search	.886	.083	.192	.118	30	11,016
You.com	.883	.370	.593	.129	31	12,020
Brave	.881	.383	.613	.167	31	12,020
Keenable	.879	**.721**	**.800**	**.510**	31	12,020
Linkup	.878	.357	.575	.130	21	7,898
Keenable realtime	.877	.709	.796	.509	31	12,020
Perplexity	.865	.585	.668	.309	30	11,636
Jina	.862	.629	.470	.030	31	11,716
Perplexity Fast	.860	.574	.666	.305	30	11,636
Claude search	.860	.427	.620	.182	27	10,400
Context	.856	.628	.516	.111	31	12,020
Exa Lite	.848	.371	.305	.219	1	420
Firecrawl	.836	.636	.472	.028	31	12,020
Google (Serper)	.834	.601	.457	.030	31	12,020
Exa	.805	.531	.462	.345	31	12,020
Tavily	.795	.279	.144	.033	31	12,020
TinyFish	.769	.563	.440	.018	31	3,472
Exa Instant	.713	.409	.374	.273	31	12,020
Octen	.660	.063	.147	.157	31	12,020
DeepSeek	.608	.183	.184	.109	12	4,820
Ceramic	.011	.000	.000	.000	31	12,007

Table 7: Scholar recall@10 by query type, 31 runs, 29 September–28 October 2026. The ceiling has 3,005 query-runs per type; the last column sums all four types. The 31 runs contain 332–420 queries each, totaling 12,020 rather than 31×420: 420 is a target, not a fixed sample size. Run counts show scored support; Exa Lite has only one run. Counts also fall after account failures. Bold marks the best engine per type (`bucket_table.py`).

D Scholar Query Types and Paired Differences

Table 7 reports every engine and its sample size, rather than only the seven lines in Figure 2. In the main window, Keenable's recall lead over Exa is .38 for 7-day papers, .10 for 30-day papers and .03 for one-year papers; it is negative in health sciences ($-.10$). The advantage is not uniform across age or domain.

For paired query-type comparisons, we retain 2,629 paper-runs from 29 runs where Keenable, its realtime mode, Kagi, Exa and ChatGPT search all have scores for all four types (10,516 query-runs). Account-side failures exclude the paper from all five; other failures score zero. We use stored identifier hits and 2,000 paired run-bootstrap resamples.

Body-keyword minus title recall is $-.154$ [$-.178$, $-.132$] for Keenable and $-.802$ [$-.833$, $-.772$] for ChatGPT search. On body keywords, Keenable's paired lead is .012 [.007, .018] over realtime, .048 [.021, .074] over Kagi, .199 [.174, .226] over Exa and .644 [.619, .669] over ChatGPT search. These pointwise 95% intervals use common papers, unlike the available-case means in Table 7; they do not measure identifier-matcher error. Full contrasts and replay inputs are in `audit/scholar_paired.json` and its companion JSONL file.

E Judge Sensitivity Checks

Audit protocol. The 200 queries in Section 3.4 come from 17–29 October runs with at least 19 engines. Each Claude rater uses a separate random order in 24 sessions of 50 items; Gemini and Grok grade one item per call at temperature 0. All use the same written guideline. Raters assign mean grades of 1.87–1.97 versus the production judge's 1.60, so agreement does not imply identical grade calibration.

Evidence length. The staged snippet-versus-page check changes 25% of 400 grades (95% interval 21–29%), but the mean grade change is only +.05 [$-.01$, +.11]. Positive and negative changes can cancel in the mean.

On 803 news results from eight queries, a Claude Opus 5.5 judge grades both full snippets and versions capped at 200 characters. The resulting engine orders agree at $\tau_b = .73$ (95% interval .40–

.84); mean absolute share change is .084. Exa and Keenable lose, while Bing (SearchAPI) and Google (Serper) gain. The sample does not distinguish the top two at either length. Snippet length is thus part of the measured API quality, not a controlled property of the underlying page.

Literal quotes. Of 1,200 rare query-runs, 504 contain quotes. On unquoted queries, Exa, Perplexity and Brave (LLM ctx) lead; Keenable is fourth. Replacing the rare score with its unquoted-only value gives overall $\tau_b = .96$ and a 3.3-point Keenable lead over Exa. Literal phrase matching therefore affects the rare ranking even when the overall order changes little. These checks use LLM judgments, not human labels.

F Finance Matching and Checks

Matching rules. Money must fall within 0.1% of the reference value; headcounts within 15%. A scale word fixes a money figure's scale. Bare figures, including parenthesized losses, can mean thousands or millions only when the preceding 200 characters contain a currency or financial cue. Years, percentages and multiples do not qualify. Diluted EPS also needs "EPS", "per share" or "earnings per". Money gets no LLM backstop; other fields can receive an upgrade, never a revoked hit. The full rules are in `src/needle/finance/canon.py`.

Why the narrow money band? An LLM audit of 399 results checks company, period and line item. Money-hit precision without table reading is .93 at the 0.1% band and .88 at 2%: the wider band admits more wrong periods and related line items. Money backstop upgrades have precision .36, which motivates excluding them. These are model judgments, not human gold labels.

Unseen-query check. A separate sample has 307 unseen queries (146 money, 161 other), tested on 17 engines. An LLM checks the first matching result for 335 table-reading hits, 100 exact-band money hits and 50 other-field hits, using at least the full 2,000-character input. Exact-band precision is .93. Of 335 table-reading hits, 235 confirm the correct figure and period (.70; 95% Wilson interval .65–.75; Wilson, 1927), 95 omit period labels and five contradict the answer. Counting unresolved hits as wrong gives precision .70; counting only contradictions gives .99. Confirmation rates differ by engine: for example, 13/43 for Firecrawl and 42/44 for Exa. Pooled precision cannot establish each engine's error rate.

Effect on reported ranks. Discounting each engine's main-window table-reading hits by its fresh unconfirmed fraction lowers finance recall by up to 6.0 points; overall ranks 1–10 stay fixed. Dropping table reading entirely lowers finance recall by 0.1–7.5 points, with finance $\tau_b = .84$ and overall .97; overall ranks 1–3 stay fixed. The check bounds one scoring choice, not all matcher error. The fresh sample has more money queries than production and lacks the non-money backstop, so it cannot replace the main table.

G Prompts

These Jinja templates are verbatim copies of the prompts in `src/needle/*/prompts/`; the code fills the braces at run time. Scholar title queries use no prompt.

Query generation. `projection.jinja`: News queries from feed items.

```
Today's date: {{ today }}

You are a search query generator for a search engine
      evaluation system. Generate a single search
     query that targets a specific news event
     happening today (within the past few hours)
     related to this recently-published article.

Requirements:
- Keyword-style query, not a natural language
     question
- 2-6 words, terse and direct
- The query MUST reference the specific event from
     the article, not just the person/team/topic in
     general (e.g. 'Lakers trade deadline' NOT just
     'Lakers')
- Include the year or date where it helps
     disambiguate
- Do NOT copy article titles or URLs verbatim
- Do NOT start the query with 'what' or 'why' -- use
      keyword style
- Always respond in English. If the article is not
     in English, respond with an English query
     anyways
- If the article's ideal answer would be the same a
     month from now (evergreen explainer, how-to,
     recipe collection, best-of list, gift guide,
     review of a stable product, historical
     retrospective, opinion piece), respond with
     exactly NO_NEWS_EVENT

Source kind: {{ source_kind }}
Title: {{ title }}
Summary: {{ summary }}
URL: {{ url }}

Respond with ONLY the query string (or NO_NEWS_EVENT
     ). No quotes, no explanation, no JSON.
```

`trends_projection.jinja`: News queries from Google Trends items.

```
Today's date: {{ today }}

You are a search query generator for a search engine
      evaluation system. Generate a single search
     query that targets a specific news event
     happening today (within the past few hours)
     related to this trending topic.
```

```
Requirements:
- Keyword-style query, not a natural language
    question
- 2-6 words, terse and direct
- The query MUST reference the specific event from
    the news articles, not just the person/team/
    topic in general (e.g. 'Lakers trade deadline'
    NOT just 'Lakers')
- Include the year or date where it helps
    disambiguate
- Do NOT copy article titles or URLs verbatim
- Do NOT start the query with 'what' or 'why' -- use
     keyword style
- Always respond in English. If the topic is not in
    English, respond with an English query anyways
- If the news articles do not describe a specific
    current event, respond with exactly
    NO_NEWS_EVENT

Trending topic: {{ topic }}

News context:
{{ news_block }}

Respond with ONLY the query string (or NO_NEWS_EVENT
    ). No quotes, no explanation, no JSON.
```

`filing_query.jinja`: Finance filing queries by description.

```
You generate a search query for a financial-search
    benchmark. The query must lead a web search
    engine to ONE specific SEC filing -- without
    ever citing its accession number.

You are given the filing company, form type, filing
    date, and an excerpt of the filing text.
    Produce ONE keyword-style search query anchored
     on a SPECIFIC, DISTINCTIVE detail from the
    excerpt: a named transaction or counterparty,
    an acquired business, a named facility or
    product, a precise unusual amount, an executive
     change, a named lawsuit or settlement.

Requirements:
- Keyword-style, 4-10 words total. Not a natural-
    language question. Do not start with what/why/
    how.
- Include the company name (or its common short form
    ) OUTSIDE the quotes.
- Wrap EXACTLY ONE distinctive span of 2-6
    consecutive words in double quotes, copied
    CHARACTER-FOR-CHARACTER from the excerpt. Pick
    wording specific to THIS filing -- not
    securities boilerplate (not "forward-looking
    statements", "pursuant to section", "securities
     and exchange commission", "quarterly report",
    "check mark").
- Do NOT include the filing's accession number, CIK,
     file number, or any URL.
- Always respond in English.
- If the excerpt is only boilerplate (cover page,
    checkboxes, table of contents) with no
    distinctive detail, respond with exactly
    NO_DISTINCT_QUERY.

Company: {{ company }}
Form: {{ form }}
Filed: {{ filed }}
Excerpt:
{{ text }}

Respond with ONLY the query string (or
    NO_DISTINCT_QUERY). No explanation, no JSON.
```

`code_query.jinja`: Legal CFR queries.

```
You generate a search query for a legal-search
    benchmark. The query must lead a web search
    engine to ONE specific section of the US Code
    of Federal Regulations -- without ever naming
    that section.

You are given the section's heading and its full
    text. Produce ONE keyword-style search query
    built from the section's SUBSTANCE: the
    specific obligation, prohibition, threshold,
    exemption, or defined term it establishes.

Requirements:
- Keyword-style, 4-10 words total. Not a natural-
    language question. Do not start with what/why/
    how.
- Wrap EXACTLY ONE distinctive span of 2-6
    consecutive words in double quotes, copied
    CHARACTER-FOR-CHARACTER from the section text (
    not the heading). Pick a span whose wording is
    regulatory and specific -- a defined term, an
    operative phrase, an unusual threshold -- not
    boilerplate that appears across the entire CFR
    (e.g. not "it shall be unlawful", "for purposes
     of this", "any person who").
- The rest of the query (outside the quotes) adds
    context words naming the subject matter, so the
     query works even on engines that ignore quotes.

- Do NOT include the section number, part number,
    title number, "CFR", "USC", "§", or the word "
    section" followed by a number. The task is to
    find the section from its content, not its
    citation.
- Do NOT copy the heading verbatim as the whole
    query.
- Always respond in English.
- If the section text is too generic or
    administrative to single it out (cross-
    references, addresses, OMB control numbers,
    severability boilerplate), respond with exactly
     NO_DISTINCT_QUERY.

Heading: {{ heading }}
Section text:
{{ text }}

Respond with ONLY the query string (or
    NO_DISTINCT_QUERY). No explanation, no JSON.
```

`body_query.jinja`: Scholar body-keyword queries.

```
You generate a search query for a scholarly-search
    benchmark. The query must be answerable ONLY by
     an engine that has indexed the full text of a
    specific paper -- not just its title and
    abstract.

You are given a paper's title, its abstract, and an
    excerpt from the body (methods, results,
    figures). Produce ONE keyword-style search
    query that targets a specific, distinctive
    detail found in the BODY but NOT stated in the
    title or abstract.

The query must be anchored on a GLOBALLY DISTINCTIVE
     term that THIS paper introduces, names, builds
    , or measures -- one that, on its own, already
    narrows the web to a handful of papers:
- a method, model, tool, software package, dataset,
    or benchmark the paper introduces or uses by
    name (e.g. "Toom's rule", "GC-MS carvacrol", "
    REDCap")
- a named study site, cohort, organism, cell line,
    reagent, or instrument
- a precise measured value that is itself unusual (e
    .g. "carvacrol 80.43%"), NOT a round number or
    a vague range

Ask yourself: would this query, typed into a
    scholarly search engine, plausibly return THIS
    paper near the top? If the query is built only
    from generic field vocabulary (e.g. "network
    depth", "layer contribution", "accuracy
    improvement", "training loss"), it will match
    thousands of papers -- reject it and either
    find a distinctive anchor or respond
    NO_DISTINCT_QUERY.

Shared configuration values are NOT distinctive:
    token counts, context-window sizes, batch sizes
    , parameter counts, learning rates, epochs, and
     similar hyperparameters recur across thousands
     of papers (e.g. "128 tokens", "128K context",
    "7B parameters", "batch size 32", "512 hidden
```

```
    dim"). Never anchor on these. Anchor on the
    specific NAME the paper gives its own artifact,
     or on a genuinely unusual measured result.

Requirements:
- Keyword-style, 3-8 words. Not a natural-language
    question. Do not start with what/why/how.
- At least one word must be a globally distinctive
    proper noun, named entity, or unusual precise
    value -- not generic jargon and not a shared
    hyperparameter/config value.
- Do NOT lead with, or build the query around, a
    percentage range, a round number, or a common
    configuration value.
- The query must NOT be answerable from the title or
     abstract alone: at least two of its content
    words must NOT appear in the title or abstract.
- Anchor on what THIS paper contributes -- NOT on
    prior work it cites and NOT on a well-known
    shared result. Specifically:
  - Do NOT use a citation or reference: no "et al",
    no author-surname strings (e.g. "Smith et al
    2024" or "Johnstone Güdel"). Cited names point
    at OTHER papers.
  - Do NOT anchor on an eponymous or textbook
    formula, theorem, law, transform, or
    distribution that predates and is shared across
     the field (e.g. "Cardano's formula", "Navier-
    Stokes", "Kermack-McKendrick", "Fourier
    transform").
- Do NOT reference the paper's own figures, tables,
    sections, theorems, or supplementary material
    by number (e.g. "Table 3", "Figure 2", "Theorem
     7.1", "Supplementary Table") -- those are
    internal pointers, not searchable content.
- Do NOT copy the title. Do NOT include the paper's
    arXiv id, DOI, or a URL.
- Always respond in English.
- If ONE distinctive multi-word span (3-6 words)
    appears VERBATIM in the body excerpt, wrap
    exactly that span in double quotes so exact-
    match engines can phrase-match it; copy it
    character-for-character from the body. Leave
    the rest of the query unquoted. If no
    distinctive verbatim span exists, use no quotes.

- If the body has no detail distinctive enough to
    single this paper out, respond with exactly
    NO_DISTINCT_QUERY.

Title: {{ title }}
Abstract: {{ abstract }}
Body excerpt:
{{ body }}

Respond with ONLY the query string (or
    NO_DISTINCT_QUERY). No explanation, no JSON.
```

`clue_query.jinja`: Scholar body-question queries.

```
You generate a search query for a scholarly-search
    benchmark. The query is a natural-language
    question a researcher might type to re-find a
    specific paper from precise factual clues.

You are given a paper's title, its abstract, and an
    excerpt from the body (methods, results,
    figures). Write ONE full-sentence question
    built from 2-3 precise, distinctive facts found
     in the BODY but NOT stated in the title or
    abstract -- the experimental setup, a named
    cohort, site, or instrument, or a specific
    measured result.

Example of the style (for a different paper): "In a
    German study of 145 cosmetic and personal care
    samples, what percentage contained NDMA and
    what was the maximum concentration found?"

Requirements:
- One question, 12-35 words, natural English prose.
    Not a keyword list.
- The clues must be precise enough that the question
     plausibly identifies THIS paper: include at
    least one named entity, specific setup detail,
    or unusual measured value from the body.
- At least two of the question's content words must
    NOT appear in the title or abstract.
- Anchor on what THIS paper did -- NOT on prior work
     it cites: no "et al", no author-surname
    strings.
- Do NOT reference the paper's own figures, tables,
    sections, theorems, or supplementary material
    by number.
- Do NOT copy the title. Do NOT include the paper's
    arXiv id, DOI, or a URL. No quotation marks.
- Always respond in English.
- If the body has no facts distinctive enough to
    single this paper out, respond with exactly
    NO_DISTINCT_QUERY.

Title: {{ title }}
Abstract: {{ abstract }}
Body excerpt:
{{ body }}

Respond with ONLY the question (or NO_DISTINCT_QUERY
    ), on a single line. No explanation, no JSON.
```

`tot_query.jinja`: Scholar tip-of-tongue queries.

```
You generate a search query for a scholarly-search
    benchmark. The query is a tip-of-the-tongue
    request: how a researcher describes a paper
    months after reading it, when the details have
    gone fuzzy.

You are given a paper's title, its abstract, and an
    excerpt from the body. Write ONE hedged, half-
    remembered description (1-2 sentences) of the
    paper's setup and main finding.

Example of the style (for a different paper): "Wasn'
    t there a paper looking for a threshold below
    which dietary cholesterol wouldn't harm
    arteries in primates, and they never found one?
     I think the good cholesterol dropped even
    though the total looked normal."

Requirements:
- 15-40 words, conversational English. Hedge it: "
    wasn't there a paper", "I think", "something
    about".
- Blur every specific: approximate quantities ("
    around 150", "a few hundred"), plain-language
    paraphrases of technical terms.
- Do NOT use any name the paper introduces or relies
     on: no method, model, dataset, tool, or
    benchmark names, no acronyms, no author names,
    no venue.
- Reuse at most two content words from the title;
    paraphrase the rest.
- No exact values with decimals, no identifiers, no
    URLs, no quotation marks, no figure/table
    references.
- The description must still be recognizable: keep
    the concrete setup (what was studied, on what,
    and roughly what was found).
- Always respond in English.
- If the abstract and body are too abstract to
    describe this way, respond with exactly
    NO_DISTINCT_QUERY.

Title: {{ title }}
Abstract: {{ abstract }}
Body excerpt:
{{ body }}

Respond with ONLY the description (or
    NO_DISTINCT_QUERY), on a single line. No
    explanation, no JSON.
```

Judging. `query_profile.jinja`: News and rare query profile, system prompt.

```
Classify a web search query for the Needs Met rating
     framework. Your classification is passed
    verbatim to every per-document judgement for
    this query, so all documents are judged against
     the same query interpretation. The sections
    below are shared with the per-document judge.
```

```
    References like 2/SM or 3+ are ratings on the
    judge's 0-4 Needs Met scale (0-1 FailsM, 2 SM,
    3 HM, 4 FullyM); they describe how the judge
    uses each classification. Be decisive.

## Query analysis

### MAIN vs AUXILIARY aspects classification

For each key aspect implied by the query (its
    objective, sub-questions, and the kind of
    source it calls for) we have 2 gradations:
- MAIN: Directly answers the query's primary
    objective. Foundational facts the user needs.
- AUX: Supporting details, statistics, context that
    enhances but doesn't define the answer.

Aspects describe what a document must cover, never
    the answer itself. Write "the identity of the
    author", not the author's name. Do not answer
    the query from your own knowledge: the judge
    scores documents against their own content, and
     a wrong precomputed answer would corrupt every
     judgement for this query.

List 1-3 MAIN aspects, each the single minimal fact
    a document must contain to answer the query. If
     one aspect would fully satisfy the user, list
    only that one. Dates, venues, candidates,
    outcomes, and secondary qualifiers are AUX
    unless the query explicitly asks for them.

### Query type classification

Query types:
{% include "query_types.jinja" %}
### Search operator extraction

Parse the query for explicit search operators. These
     are hard constraints the user typed
    deliberately. They are not optional concepts.

- Exact-phrase: any span inside double quotes "...".
     Each quoted span is a separate required string.

- site:domain = the document URL host must equal or
    be a subdomain of `domain`.

### Content archetype classification

Classify the query's expected content to determine
    staleness evaluation:
{% include "content_archetypes.jinja" %}
### Domain classification

Classify the query into one of these domains, or "
    other" if none fits:
- "health": Medical and health queries.
- "legal": Law, regulations, caselaw.
- "financial": Companies, tickers, markets, personal
     finance.
- "academic": Science, research, formulas.
- "tech": Software, hardware, troubleshooting.
- "news": Current events and reporting.
- "commerce": Products and shopping.
- "jobs": Job listings and hiring.

## Output

Respond with ONLY a YAML block. No text before or
    after.

'''yaml
query_type: <broad|specific|list>
query_domain: <health|legal|financial|academic|tech|
    news|commerce|jobs|other>
query_search_operators_exist: <yes|no>
query_main_aspects:
  - "<MAIN aspect>"
query_aux_aspects:
  - "<AUX aspect>"
query_content_archetype: <ephemeral|persistent|
    evergreen>
'''
```

`query_profile_input.jinja`: Query profile, user message.

```
Today's date: {{ today }}

Query: {{ query }}
```

`judgement.jinja`: News and rare relevance judge, system prompt.

```
Evaluate how well a search result document matches a
     user's query intent using the Needs Met rating
     framework (0-4 scale).

You are provided with:
- Today's date
- Query: The user's search query
- Query Analysis: precomputed classification (query
    type, domain, search-operators flag, MAIN and
    AUX aspects, content archetype)
- Document: Title, URL, optional published date,
    snippet

Adopt the Query Analysis block as ground truth and
    do not re-classify the query. Step 1 below
    defines its fields. If the block is absent,
    infer these fields from the query text yourself.
     The Query Analysis defines what to look for;
    the document itself is the only source of
    factual answers -- judge facts against the page
    's own text and the source's authority, never
    against your own knowledge of the answer. You
    may still use general knowledge to recognize
    that two names or descriptions refer to the
    same entity.

## Rating Scale

- 4 (FullyM): All MAIN aspects covered with
    substantial depth. Authoritative, comprehensive
    , directly answers the query's primary
    objective. One-stop resource.
- 3 (HM): Major MAIN aspects covered with good depth
    . Minor gaps in auxiliary details or slight
    limitations, but the reader's core need is
    still substantially satisfied.
- 2 (SM): Partial MAIN coverage, or only passing/
    superficial mention without actionable depth.
    May touch the general topic while missing the
    precise query elements. Trips at least one
    document condition.
- 1 (FailsM): Marginal relevance only. Topic-
    adjacent but fundamentally misses the query's
    relevance or provides negligible useful
    information.
- 0 (FailsM): Off-topic, broken, inaccessible,
    duplicate URL, or wholly fails the query.

Rating and label must always be consistent:
- Rating 0 = FailsM
- Rating 1 = FailsM
- Rating 2 = SM
- Rating 3 = HM
- Rating 4 = FullyM

If there is any mismatch, correct it before
    outputting.

## Step 1: Query analysis

### MAIN vs AUXILIARY aspects

For each key aspect implied by the query (its
    objective, sub-questions, and the kind of
    source it calls for) we have 2 gradations:
- MAIN: Directly answers the query's primary
    objective. Foundational facts the user needs.
- AUX: Supporting details, statistics, context that
    enhances but doesn't define the answer.

Missing AUX alone never drops a document below 3/HM
    if all MAIN aspects are adequately covered.
Additionally, queries might have hard caps that
    limit the maximum possible label.

### Query types

The query type sets how strict the content
    requirements are.
```

```
Query types:
{% include "query_types.jinja" %}
### Search operators

Parse the query for explicit search operators. These are hard constraints the user typed deliberately. They are not optional concepts.

- Exact-phrase: any span inside double quotes "..." must appear VERBATIM in the snippet (the content between `<<<SNIPPET>>>` and `<<</SNIPPET>>>`) or the document title (case-insensitive, whitespace-normalized; ignore surrounding punctuation). A match in the URL, breadcrumb, navigation links, or any other metadata does not count and never satisfies the operator. Do not assume or infer presence: locate the exact span in the snippet or title, or treat it as missing. Each quoted span is a separate required string.
- site:domain = the document URL host must equal or be a subdomain of `domain`.

When a query carries no operators, skip the operator gates and score normally.

### Content archetypes

The content archetype determines staleness evaluation:
{% include "content_archetypes.jinja" %}
## Step 2: Document analysis

### Document integrity check

Before any relevance analysis, confirm the page is reachable and functional:
- "fail": Fails integrity (rate 0). Error pages (404, 500), blank/metadata-only pages, hard paywalls blocking all content, pages with only boilerplate/navigation and no substantive content.
- "limit": Passes with limitation. Social media previews, academic abstracts behind paywalls -- evaluate visible content, note limitation in reasoning. Cap at 2/SM when the whole page is required but unavailable.
- "pass": Passes completely. Content is informative.

### Document conditions

Three conditions constrain the maximum rating. All applicable conditions must hold for the document to qualify for 3/HM or higher. A condition can either pass, fail, or be not applicable.

Relevance condition (all query types):
- The page has to match the right topic, entity, jurisdiction, and domain context.
- Wrong entity = 0/FailsM no matter how good the content is (e.g., a Domino's page for a "Sal's Pizzeria Brooklyn" query).
- An under-specified reference to the same entity is NOT a wrong entity: a document that omits a middle name, an alias, or leaves a counterparty unnamed still passes the relevance condition when nothing contradicts the query's entity.
- Wrong jurisdiction = 0/FailsM for legal/financial queries.
- General topic match without specific query elements = fails the relevance condition for "specific" queries.

Coverage condition (all query types):
- MAIN aspects have to appear with real depth, not bare keywords or passing mentions.
- "Coming soon" without a date != adequate for a "when does X release" query.
- A title mention without explanation != adequate coverage.
- "broad" queries: At least one MAIN aspect with meaningful depth.
- "specific" queries: The specific requested answer must be present.
- "list" queries: ≥50% of requested items/aspects must be covered.
- A page that credibly overturns the query's premise still answers it: for "did X die", a report showing that X is alive or that the event belongs to a related entity satisfies coverage. Do not demand confirmation of a premise the document credibly refutes.
- A secondary query detail the document leaves unconfirmed (a date, a middle name, an unnamed counterparty) is a minor gap to note in reasoning, not a coverage failure, when the queried fact itself is reported.
- If condition fails, document is capped at 2/SM.

Completeness condition (for "list" queries):
- Covering <50% of requested items = capped at 2/SM.
- Covering ≥50% but <100% = eligible for 3/HM.
- Covering ~100% with depth = eligible for 4/FullyM.

Gate logic summary:
- All conditions pass -> eligible for 3/HM or 4/FullyM based on depth and authority
- Any condition fails -> maximum 2/SM (or 0/FailsM for relevance condition failures)
- See the section below for additional constraints for specific, dated current-event queries.

### Current event freshness gate

When the query refers to a specific, dated current event (e.g., "Medvedev vs Berrettini Monte-Carlo Masters Round of 32, April 8, 2026"), these constraints override the standard 2/SM cap. Apply strictly: topical overlap is not sufficient.

- Entity without event: 1/FailsM maximum. The document discusses the correct entity but does NOT address the specific event the query names. No exceptions. This includes: Wikipedia biographies, player/team profile pages, career statistics pages, game logs missing the requested date, generic relationship or biography articles, stats dashboards without the specific game, and any page that covers the entity generally without referencing the event itself.
- Wrong event: 1/FailsM maximum. The document covers a DIFFERENT event involving the same entity. Use the query to distinguish: a 1st-leg match article is wrong-event for a 2nd-leg query; a March committee markup article is wrong-event for an April 8 debate query; a 2024 Tommy John article is wrong-event for a 2026 shoulder injury query.
- Pre-event for result queries: 1/FailsM maximum. When the query seeks a result, outcome, or post-event status, any preview, prediction, or pre-event article = 1/FailsM. This includes odds/previews for completed games, pre-match analyses for result queries, and pre-announcement speculation for outcome queries.
- How to verify: Before rating 2+ for a specific dated-event query, confirm the document contains at least one CONCRETE detail matching the queried event (the specific date, score, quote, ruling, or named development). General coverage of the entity or topic without a concrete detail from the event = 1/FailsM.
- Scope: This gate only fires for specific, dated current-event queries. Otherwise, the standard max 2/SM for coverage condition failures remains unchanged.

### Staleness and temporal fit

Based on the expected content archetype, apply staleness checks:

"ephemeral" content:
- Strict temporal alignment is required. A page about 2024 performers does not serve a "2025 performers" query.
- Breaking news: 0-48 hours = recency paramount. 2-7 days = balance with comprehensiveness.
- Future-tense queries accept predictions/previews. Past-tense queries require actual results: a
```

```
    write-up published ahead of the event cannot
    say "who won".
- Recurring annual events: Previous year's info is
    acceptable until next year's is announced.

"persistent" content:
- Obsolete technology/platforms (Windows Phone,
    defunct services) = stale.
- Outdated business info (old CEO, wrong pricing,
    old contact details) = stale.
- Stale "Just launched!" banners for features that
    shipped as standard long ago = stale.

"evergreen" content:
- A recent publication date by itself doesn't make
    it stale. Only superseded factual developments,
     withdrawn methodologies, or incomplete
    timelines qualify.
- A 2016 write-up covering a settled scientific
    principle beats a shallow 2025 one.

Staleness penalties:
- Critical mismatch (wrong year event, wrong entity
    status) = 0/FailsM
- Severe (multiple outdated facts, functionally
    broken purpose) = downgrade 2 levels
- Moderate (some outdated details, minor temporal
    issues) = downgrade 1 level
- Minor/peripheral (old copyright year, minor detail
    ) = note in reasoning only

### Additional rules

Apply these principles to avoid over-penalizing high
    -quality documents:
- When a high-trust source (official government site
    , ESPN, an established news outlet) reports
    something different from what the query seems
    to expect, believe the page. Dock it only if it
     is clearly stale or a weak-authority source
    that contradicts several authorities.
- The query's criteria capture ideas, not literal
    keywords. Judge whether the page satisfies the
    underlying information need, not whether
    particular keywords appear. "battery-electric
    car" = "EV". "headcount reduction" = "layoffs".
     Give equivalent ideas full marks. The only
    exception are the spans the user placed in
    double quotes or other explicit operators,
    which must be matched literally and are not
    subject to conceptual substitution.
- For live or dynamic data (stock quotes, weather,
    sports scores), small gaps from expected values
     are fine when the source is authoritative.
    Dock only large deviations that signal
    staleness.
- Examples implied by the query are illustrative,
    not mandatory. A document covering equivalent
    examples satisfies the criterion.

### Domain-specific hard caps

Apply when the query falls into these domains:

- "health": Authority is paramount (CDC/WHO > blogs)
    . One dangerous error (unproven therapy) = 0/
    FailsM. Prioritize peer-reviewed sources and
    official guidelines.
- "legal": Jurisdiction must match. Mismatch = 0/
    FailsM. A critical mistake (wrong fees, expired
     deadlines) caps at 2/SM.
- "financial": Verify the entity (right company/
    ticker/exchange) - required. Wrong entity = 0/
    FailsM. Require concrete quantitative data, not
     vague claims.
- "academic": Core formula/principle errors = 1-2.
    Peripheral citation errors don't drop below 3/
    HM if core is correct.
- "tech": Safety notes required for risky procedures
     (editing the registry, taking hardware apart).
     Crowd-verified fixes can beat terse official
    pages.
- "news": Critical factual errors = 1-2. Separate
    reporting from speculation. Prefer established
    outlets.
- "commerce": Require an exact product match (right
    model/version/color/size). Wrong variant with
    same brand = 2/SM max. Different product line =
     0/FailsM.
- "jobs": If query specifies work modality (remote/
    telemedicine), listings must match. On-site
    listings for remote queries = significant
    downgrade.

### Search operator compliance gate (applies only
    when query has search operators)

Operators are explicit user constraints, equal in
    force to entity/jurisdiction:
- Required exact "phrase" absent from both the
    snippet and the title = 0/FailsM. A match found
     only in the URL, breadcrumb, navigation, or
    metadata does NOT satisfy the operator.
- site: host mismatch = 0/FailsM

A document may be perfectly on-topic and still fail
    this gate.

## Step 3: Output

Respond with only a YAML block. No text before or
    after.

'''yaml
reasoning: "<2-3 sentences. State the query type (
    broad/specific/list), which document conditions
     passed/failed, key evidence from the document,
     and any penalties applied (staleness, domain
    gate). Be specific about WHAT content was
    present or missing.>"
document_integrity_check: <fail|limit|pass>
document_relevance_condition: <fail|pass>
document_coverage_condition: <fail|pass>
document_completeness_condition: <fail|pass|na>
freshness_gate: <fail|pass|na>
staleness_gate: <fail|pass>
domain_gate: <fail|pass|na>
search_operators_gate: <fail|pass|na>
rating: <0-4>
label: <FailsM|SM|HM|FullyM>
'''

Gate field rules:
- document_relevance_condition and
    document_coverage_condition apply to every
    query: always pass or fail.
- document_completeness_condition: na unless the
    query type is "list".
- freshness_gate: na unless the query targets a
    specific, dated current event.
- staleness_gate: fail only when a staleness penalty
     applies; pass otherwise.
- domain_gate: na when the query domain is "other".
- search_operators_gate: na when the query has no
    search operators.

Output rules:
- YAML only. No markdown fences, no commentary, no
    extra fields.
- `reasoning` must be 2-3 sentences grounded in
    document evidence.
- Rating and label must be consistent (see rule
    above).
```

`judgement_input.jinja`: Judge, user message.

```
Today's date: {{ today }}

Query: {{ query }}

{% if profile -%}
Query Analysis (precomputed -- adopt as ground truth
    , do not re-classify):
- Query type: {{ profile.query_type }}
- Domain: {{ profile.query_domain }}
- Search operators exist: {{ "yes" if profile.
    query_search_operators_exist else "no" }}
- Content archetype: {{ profile.
    query_content_archetype }}
{% if profile.query_main_aspects -%}
- MAIN aspects:
{% for aspect in profile.query_main_aspects -%}
{{ "  - " ~ aspect }}
```

```
{% endfor -%}
{% endif -%}
{% if profile.query_aux_aspects -%}
- AUX aspects:
{% for aspect in profile.query_aux_aspects -%}
{{ "  - " ~ aspect }}
{% endfor -%}
{% endif %}
{% endif -%}
Document to evaluate:
- Title: {{ title }}
- URL: {{ url }}
{% if published -%}
- Published: {{ published }}
{% endif -%}
{% if content -%}
- Page Snippet:
<<<SNIPPET>>>
{{ content }}
<<</SNIPPET>>>
{% endif -%}
```

`query_types.jinja`: Included by the profile and judge prompts.

```
- "broad": General or exploratory topic queries ("
     what is inflation", "zendaya", "origins of jazz
     ").
  - MAIN = foundational defining facts. AUX =
     statistics, supplementary details.
  - The page has to cover at least one main aspect
     in real depth to reach 3+.
- "specific": Queries after a single closed answer
     ("when X launches", "who won the final", "how
     much Y is worth").
  - MAIN = the specific requested fact. Nothing else
      substitutes.
  - Hard cap: no specific answer present = capped at
      2/SM, regardless of how much related context
     is provided. A page that credibly overturns the
      query's false premise counts as providing the
     answer.
- "list": Comparisons, multi-item requests, pros-and
     -cons ("A vs B", "best 10 albums", "upsides and
      downsides of Z").
  - MAIN = coverage of all requested items/aspects.
  - Hard cap: Covering <50% of requested items =
     capped at 2/SM.
```

`content_archetypes.jinja`: Included by the profile and judge prompts.

```
- "ephemeral": Sports scores, weather, breaking news
     , stock prices, event schedules. Value decays
     rapidly.
- "persistent": Official websites, product pages,
     how-to guides for software. URL always relevant
      but content freshness matters.
- "evergreen": Historical facts, scientific
     principles, definitions, recipes. Largely time-
     insensitive.
```

`answer_match.jinja`: Finance LLM backstop for non-money fields.

```
You are verifying search results for a company fact-
     lookup benchmark.

Search query: {{ query }}
Fact being looked up: {{ field }}
Gold answer: {{ value }}
{% if aliases -%}
Also acceptable forms: {{ aliases | join("; ") }}
{% endif %}
Search result:
- Title: {{ title }}
- URL: {{ url }}
- Snippet: {{ snippet }}

Does this search result state the gold answer? Any
     equivalent form counts --
different wording, formatting, units, reasonable
     rounding (e.g. "$416.2B" for
416161000000), nicknames, or the answer appearing in
      the result URL. The result
must actually contain the fact; being about the
     right company or topic is not
enough. If the result states a clearly different
     value, answer no.

Answer with a single word: yes or no.
```